Chapter 3 Student Notes and Labs

Neuromodulation

Student Notes

What this chapter is for

Chapters 1 and 2 gave you a neuron that integrates inputs and sometimes fires. Chapter 3 answers a question that model cannot answer on its own: given that neurons are firing all the time, when does the brain actually learn, and what decides which synapses change?

The chapter's answer is that fast electrical signaling and learning are coupled by a chemical layer called neuromodulation. A handful of small brainstem and basal forebrain nuclei broadcast chemicals across most of the brain. Those chemicals do not carry sensory content. They carry context: safe or threatened, better or worse than expected, stable or changed. Pre- and postsynaptic activity alone can induce Hebbian potentiation and depression (Chapter 5), but in the intact brain much of that plasticity is gated and scaled by this context, which decides whether, how strongly, and in which direction a change is expressed and consolidated.

If you are coming from machine learning, this chapter is designed to correct three instincts:
1) assuming the decision about when to learn is imposed from outside, by a switch between training and deployment,
2) assuming an error signal can always be traced back exactly to the parameter that caused it, and
3) reading "dopamine" as "reward."

Your goal is to be able to explain, in plain language, why the brain does not run exact backpropagation as deep networks do (how it approximates credit assignment is an open question), what the neuromodulatory systems contribute instead, and why that approach is a reasonable design under uncertainty rather than a defect.

Two timescales, one circuit

Synaptic transmission is fast (milliseconds), spatially precise, and short-lived. It moves information.

Neuromodulation is spatially diffuse and slow in its action. The signals themselves can be fast: a dopamine neuron's burst at a surprise lasts about a tenth of a second, and locus coeruleus neurons respond to a salient stimulus with a similar latency. What is slow is the effect. Most neuromodulator receptors are G-protein-coupled receptors that set off chains of chemical reactions inside the cell, so their effects on excitability and synaptic strength unfold over seconds to minutes and sometimes longer. Neuromodulation does not carry detailed sensory content. It changes how circuits respond to information: it raises or lowers excitability, biases competition between pathways, and decides whether, how strongly, and in which direction synapses change (Figure 3.2 in the chapter).

The same wiring therefore behaves differently depending on chemical state. This is why the "rigid machine" picture from the start of the chapter fails: identical inputs do not produce identical outcomes, because the circuit is never in exactly the same chemical state twice.

You should be able to say: synapses carry the message; neuromodulators set the terms on which the message is received.

Volume transmission versus synaptic transmission

At a classical synapse, a fast transmitter such as glutamate or GABA crosses a cleft about 20 nanometers wide, acts on one postsynaptic partner, and is cleared within milliseconds.

Neuromodulators are often released from varicosities along the axon rather than at a tight synapse. They diffuse through extracellular space over micrometers and reach receptors on many cells and synapses at once, with effects that last seconds to minutes. This is called volume transmission (Figure 3.1 in the chapter).

The engineering consequence: a neuromodulator cannot address a single synapse. It addresses a region. Any precision in the resulting learning has to come from somewhere else (see eligibility traces below).

The "-ergic" vocabulary

Neurons are often named by the chemical they release. The suffix "-ergic" means using or releasing.
- dopaminergic: releases dopamine
- noradrenergic: releases norepinephrine (noradrenaline)
- serotonergic: releases serotonin
- cholinergic: releases acetylcholine

These labels describe chemical influence, not meaning. A dopaminergic neuron does not "represent reward"; it releases a chemical that other circuits interpret in context.

Why the systems are diffuse

Neuromodulatory nuclei contain very few cells (Figure 3.3 in the chapter maps the four systems). The human nucleus basalis of Meynert holds roughly 200,000 cholinergic neurons per hemisphere, and the whole basal forebrain cholinergic system is a few hundred thousand cells, against about 16 billion cortical neurons. One cell compensates with the size of its axon: Matsuda and colleagues (2009) found that a single rat nigrostriatal dopamine neuron's arbor is about half a meter long and covers on average 2.7 percent of the striatum's volume (up to about 6 percent), with release sites estimated at several hundred thousand per neuron in the rat and more than a million in the human. One cell can speak to hundreds of thousands of synapses at once.

To an engineer this looks wasteful. It is not. The signal being sent is global by nature. "Outcomes are better than expected" or "the environment just became unpredictable" is information every circuit needs at once. Broadcasting it from a small source is cheaper and more coherent than computing it locally in millions of places. Diffuse projection lets a few hundred thousand neurons coordinate billions without a central controller.

Dopamine neurons signal reward prediction error, not reward

The chapter's most important correction comes from Wolfram Schultz's recordings of single midbrain dopamine neurons in monkeys working for juice: a large population of dopamine neurons signals a reward prediction error, the difference between the reward received and the reward predicted (Figure 3.5 in the chapter).

- Outcome better than expected: a burst above baseline.
- Outcome exactly as expected: firing stays near baseline.
- Outcome worse than expected (predicted juice withheld): a dip below baseline, timed to when the reward was due.

Two consequences follow. First, a fully predicted reward produces almost no response, even though the reward is still pleasant. Second, once a cue reliably predicts a reward, the burst moves earlier in time, to the cue. Schultz, Dayan, and Montague (1997) showed that this is what the error term of Sutton's temporal-difference learning looks like as learning proceeds (Figure 3.6 in the chapter). The causal test came later: optogenetically driving dopamine neurons when a reward was already predicted made rats learn about a cue they would otherwise have ignored (Steinberg et al. 2013).

Hold the account with the chapter's caution. It describes the phasic response of a large population of dopamine neurons, not every dopamine neuron or everything dopamine does. Dopamine neurons are heterogeneous: some respond to aversive events, novelty, or salience, and some carry signals tied to movement and position, and what the slow striatal dopamine ramps mean is an active debate. Note also that this reward prediction error is a single number about value. Chapter 6 introduces a sensory prediction error, which is high-dimensional and about content; the two share a name and a logic but are different signals.

The dopamine signal says when to learn and in which direction (bursts favor potentiation, dips the opposite change, at striatal synapses carrying different dopamine receptors). It does not say which synapses should change: it is one scalar broadcast to the entire striatum and much of cortex.

The credit assignment problem

When an action succeeds or fails, which internal components deserve the credit or blame?

In an artificial network this is solved by construction: the architecture is layered, the objective is explicit, the correct answer is supplied, and backpropagation computes each weight's exact contribution to the error.

The brain has none of those conditions. It receives no labels. Outcomes are delayed, noisy, and ambiguous. Circuits are recurrent and overlapping, and each neuron takes part in many functions at once. By the time an outcome is known, the activity that caused it has been overwritten.

Izhikevich called this last difficulty the distal reward problem. And because thousands of synapses usually contribute and none is decisive, the question "which synapse caused this outcome?" often has no single answer.

The brain has no known way to do what backpropagation does exactly, and whether and how it approximates it is an open question. Crick (1989) objected that nothing in cortex looked like a way to

send each synapse its share of a global error. The objection has weakened without disappearing: random feedback weights are enough for a network to learn, dendritic bursts in pyramidal cells can carry an error signal separate from the forward signal, and Lillicrap and colleagues (2020) argue the brain may approximate backpropagation-like learning through its abundant feedback connections. Chapters 5 and 6 return to this. The chapter's narrower point: whatever the brain does about credit assignment, it is not the clean procedure of the artificial case, and much of what it does instead is visible in the neuromodulatory systems.

Eligibility traces: the physical workaround

If a global signal cannot identify the responsible synapse, how does any specificity arise?

Each synapse keeps a short-lived chemical memory of its own recent activity. When a synapse is used, molecular processes put it into a transient modifiable state, an eligibility trace, which decays over seconds to minutes. If a neuromodulatory signal arrives while the trace is still present, the synapse changes. If the signal arrives after the trace has faded, nothing happens (Figure 3.7 in the chapter). The windows have been measured: in striatal slices, a spine enlarged only when dopamine arrived about 0.3 to 2 seconds after the activity (Yagishita et al. 2014); in hippocampal slices, dopamine applied within about a minute after a pairing turned depression into potentiation, but not after ten minutes (Brzosko et al. 2015).

This produces what the chapter calls a three-factor learning rule (written out in Figure 3.8 of the chapter):

change in strength = (learning rate) x (eligibility trace at the moment the signal arrives) x (prediction error)

Factors one and two (presynaptic and postsynaptic activity, Hebb's factors, combined into the trace) are local to the synapse and say "I was recently involved." Factor three, the neuromodulator, is global and says "something significant just happened." Within this rule neither is enough alone, and the sign of the pairing and the sign of the signal together set the direction of change. (Remember the caveat from the notes above: Hebbian plasticity can be induced by activity alone in slices; the three-factor rule describes how neuromodulators gate and scale much of it.) The trace is the brain's answer to the temporal gap between action and outcome, and it supplies the locality that volume transmission cannot. Chapter 5 describes its molecular form, synaptic tagging and capture.

Bias, and local rather than global objectives

Each update is small and probabilistic. A synapse that was active by coincidence just before a reward will be strengthened a little too, and nothing prevents that. Over many trials, however, synapses that genuinely tend to precede success accumulate more strengthening than synapses that were merely present.

The result is a bias: a statistical tendency toward activity patterns that have tended to work, which is Thorndike's law of effect (1911) with a mechanism. The chapter is careful with the word. Bias here does not mean error; it means the gradual shaping of probability under uncertainty.

The chapter is equally careful about loss functions. No single global objective is handed to the brain and minimized by gradient descent. But many circuits are well described as minimizing a local error or cost: the temporal-difference error that dopamine signals is one, and Chapter 6 adds sensory prediction error and near-optimal cue combination. The accurate statement is that the brain has many local objectives and no global one, and neuromodulators are among the signals that define them.

The other neuromodulators regulate how, not what

Dopamine is the best understood system because reward is easy to manipulate in experiments. The chapter's section on norepinephrine, serotonin, and acetylcholine ties each of the others to a context of its own:

- Norepinephrine (locus coeruleus): surprise and arousal. Aston-Jones and Cohen proposed a phasic mode, in which bursts sharpen responses to task-relevant events, and a tonic mode, in which a high baseline favors disengaging and exploring. In Yu and Dayan's account, norepinephrine reports unexpected uncertainty (the world has changed), and acetylcholine reports expected uncertainty (the world is noisy but stable).
- Serotonin (raphe nuclei): the least settled. It has been tied to patience, aversion, behavioral inhibition, and the regulation of plasticity, and no unification has succeeded. One concrete result: dorsal raphe serotonin neurons in rats raised their firing while the rats waited for a delayed reward and dropped it when they gave up (Miyazaki, Miyazaki, and Doya 2011).
- Acetylcholine (basal forebrain and brainstem): the balance between taking in new information and using old. Hasselmo showed it suppresses the recurrent synapses that complete stored patterns while sparing incoming inputs, so high acetylcholine (attentive waking) favors encoding and low acetylcholine (slow-wave sleep) favors recall and replay.

Together they regulate the mode of processing rather than its content: how strongly to respond, how much to trust the current model, whether to persist or switch, and whether to write or to read.

Reference: the four major neuromodulatory systems

The chapter maps the systems in Figure 3.3 and the cholinergic system on its own in Figure 3.4. Use this table for Lab 3.5 and as a lookup when later chapters refer to a nucleus by name.

System	Source nuclei (location)	Main targets	Context it signals	Typical timescale
Dopamine (dopaminergic)	Ventral tegmental area and substantia nigra pars compacta (midbrain)	Striatum, nucleus accumbens, prefrontal cortex	Reward prediction error, signaled by a large population of neurons; subpopulations also respond to novelty, salience, aversive events, and movement	Bursts of about 100 ms and dips of a few hundred ms on a slowly varying tonic level
Norepinephrine	Locus coeruleus	Almost the entire	Arousal and	Phasic responses

System	Source nuclei (location)	Main targets	Context it signals	Typical timescale
(noradrenergic)	(pons)	cortex, hippocampus, thalamus, cerebellum	surprise; unexpected uncertainty (the world has changed); phasic and tonic modes of cortical gain	of hundreds of ms; tonic level shifts over minutes
Serotonin (serotonergic)	Raphe nuclei, chiefly the dorsal and median raphe (brainstem)	Widespread cortex, hippocampus, striatum, amygdala	Least settled: patience and waiting, aversion, behavioral inhibition, regulation of plasticity	Seconds to minutes; among the slowest modulatory signals
Acetylcholine (cholinergic)	Nucleus basalis of Meynert (to all of neocortex; roughly 200,000 cholinergic neurons per hemisphere in humans), medial septum (to hippocampus), pedunculopontine and laterodorsal tegmental nuclei of the brainstem (to thalamus)	Cortex, hippocampus, thalamus	Expected uncertainty; the balance between encoding new information (high acetylcholine) and recall (low acetylcholine)	Seconds, with faster local release in cortex

Scale, for reference: the whole basal forebrain cholinergic system is a few hundred thousand cells against about 16 billion cortical neurons, and a single rat nigrostriatal dopamine axon covers on average about 2.7 percent of striatal volume (up to about 6 percent).

Two cautions. First, every one of these chemicals is also used in fast, point-to-point synapses somewhere in the nervous system; the table describes their modulatory role, which is the chapter's subject. Second, the "context" column is a summary of current understanding, not a settled fact. Dopamine neurons are heterogeneous, and serotonin in particular has several proposed roles that no account has yet unified; the chapter offers stability (holding a course of action) only as one account.

What physically changes

A synapse is a molecular machine, not a gap. Learning can change receptor density, neurotransmitter release probability, the protein scaffolding that holds the synapse together, the shape of the dendritic

spine, and gene expression in the cell. “Memory” is a change in structure and in probability, carried forward because structure persists.

Neuromodulation as the gatekeeper of plasticity

If synapses changed every time neurons fired, the brain would be unstable: every stray coincidence would be learned and every memory would be overwritten by ongoing activity (the stability-plasticity dilemma, to which Chapter 5 returns). Gating much of learning on chemical context is the chapter’s candidate design principle for biological intelligence. The evidence includes depletion experiments: kittens whose visual cortex lacked both norepinephrine and acetylcholine failed to rewire after one eye was closed, and in adult rats pairing a tone with nucleus basalis or ventral tegmental area stimulation expanded the tone’s share of auditory cortex.

Clinical evidence

The chapter’s rewritten Clinical Evidence section rests on cases where neuromodulatory tone has been altered in humans:
- Parkinson’s disease: Ehringer and Hornykiewicz (1960) found striatal dopamine almost absent; L-DOPA (Carlsson’s animal work in 1957, Cotzias’s oral dosing in 1967) remains the standard treatment. Frank, Seeberger, and O’Reilly (2004) showed that patients off medication learned better from bad outcomes and on medication better from good ones: same patients, same task, different dopamine tone.
- Antipsychotics block dopamine receptors; their clinical dose tracks binding to what is now called the D2 receptor (Creese, Burt, and Snyder 1976), and their parkinsonian side effects are dopamine loss produced by pharmacology.
- Alzheimer’s disease: the nucleus basalis loses most of its neurons (Whitehouse et al. 1982). Cholinesterase inhibitors raise acetylcholine levels and give a modest but consistent cognitive benefit at six months (Cochrane review), as expected from raising tone without restoring lost cells.
- Deep brain stimulation (Figure 3.9 in the chapter) relieves Parkinson’s symptoms but is electrical: it acts on circuit activity, not on transmitter levels, and how it works is still debated. It is “neuromodulation” only in the clinical sense of the word, not the chapter’s.

Implications for artificial intelligence

The chapter draws four lessons for AI design:
1) The brain separates activity from learning from the inside. In an artificial network a training/deployment switch imposed from outside makes that separation; the brain runs and learns continuously and decides internally, from neuromodulatory context, whether a moment of activity may change the wiring. A system that keeps learning after deployment needs something of this kind.
2) Learning rules themselves can depend on state: uncertainty, novelty, and context modulate how much the system updates (Doya’s 2002 mapping of the four systems onto error signal, learning rate, exploration, and discounting).
3) Approximate credit assignment can be sufficient. Networks trained with random feedback weights learn almost as well as with exact backpropagation, and the brain broadcasts a single number to a whole region and still learns.
4) Neuromodulation reframes the objective. No single loss function is handed to the brain; it reduces

many local errors, the temporal-difference error among them, under the constraint that it must remain viable while it learns. Success is defined operationally.

Practical summary

- Fast synaptic signaling moves information; neuromodulation, fast to signal but slow to act, sets the chemical context in which that information is processed.
- Neuromodulatory nuclei are small and project diffusely because the signal they carry is global by nature.
- A large population of dopamine neurons signals a reward prediction error (the population is heterogeneous). Predicted rewards produce no response; the response migrates to the earliest predictive cue.
- The brain does not run exact backpropagation; how it approximates credit assignment is an open question, and neuromodulators are a large part of what it does instead.
- Eligibility traces plus a global modulatory signal form a three-factor rule that bridges the gap between action and outcome.
- Updates are small and probabilistic; learning is the accumulation of bias. There is no single global loss, but many circuits minimize local errors such as the temporal-difference error.
- Much plasticity is gated: activity alone can induce Hebbian change, but neuromodulators decide whether, how strongly, and in which direction it is expressed and consolidated.
- Norepinephrine signals surprise and unexpected uncertainty, serotonin is the least settled (patience, aversion, inhibition), and acetylcholine sets the balance between encoding and recall.

Labs

Lab Notes: how to succeed in these labs

These labs continue the approach of Chapter 2: you will treat learning as a constrained dynamical process and simulate it in simplified form with pencil, paper, dice, and small calculations. Where a lab offers an optional code extension, it is deliberately short; the point is to see the mechanism, not to build a simulator.

General rules:
1) Work in steps. Write intermediate values.
2) Draw the system before computing.
3) Finish every lab with an interpretation: what changes if the parameters change, and what does the biology gain or lose?

What to hand in for every lab

- A clean diagram (a photo of a hand drawing is fine).
- A calculation table.
- A short interpretation (6-12 sentences).

Lab 3.1 - Reward Prediction Error by Hand (why dopamine moves to the cue)

Purpose

To reproduce, numerically, the two signature findings in Figures 3.5 and 3.6 of the chapter: the dopamine response to a reward shrinks as the reward becomes predictable, and the response reappears at the cue that predicts it.

Materials

Notebook, pen or pencil, calculator (recommended).

Given parameters (use exactly these)

Prediction V (how much reward the cue predicts) starts at V = 0.
Reward on every trial: r = 1.
Learning rate: alpha = 0.25.
Prediction error at the reward: delta = r - V.
Update after each trial: V_new = V + alpha * delta.
Treat the dopamine response at the reward as delta, and the dopamine response at the cue as V (the amount of reward the cue now predicts).

Setup diagram (do this first)

1) Draw a timeline for one trial: a cue at t = 0 s, a reward at t = 2 s.
2) Above the timeline, leave space for two bars: “dopamine at cue” and “dopamine at reward.”

Procedure (follow in order)

Part A - Learning the prediction

1) Make a table with columns: trial, V before, delta at reward, V after, dopamine at cue.
2) Trial 1: V = 0, so delta = 1 - 0 = 1.0. V after = 0 + 0.25 * 1.0 = 0.25. Dopamine at cue = 0.
3) Trial 2: delta = 1 - 0.25 = 0.75. V after = 0.4375. Dopamine at cue = 0.25.
4) Continue through trial 8. You should find V after trial 8 is about 0.90 and delta at the reward has fallen to about 0.13.
5) Draw the two bars for trial 1 and for trial 8 on your diagram.

Part B - Omission

6) On trial 9 the cue appears but no reward is delivered (r = 0).
7) Compute delta = 0 - V. Record its sign and size.
8) Update V as usual and note what happened to the prediction.

Part C - Sensitivity check

9) Repeat Part A with alpha = 0.05 for 8 trials. How far does V get?
10) Repeat Part A with alpha = 0.9 for 4 trials. What does the sequence of delta values look like?
11) Write 6-10 sentences: why does a reliable reward become “invisible” to dopamine, where did the signal go, and what does the omission dip tell the rest of the brain?

What to record

Table for Parts A through C, the two-bar diagrams for trial 1 and trial 8, and your interpretation.

Lab 3.2 - Eligibility Traces and the Temporal Gap (three-factor rule)

Purpose

To see why a global, late-arriving signal can still produce synapse-specific learning, and what is lost when the delay between activity and outcome grows.

Given parameters

When a synapse is active, its eligibility trace is set to e = 1.0.
The trace halves every 1 s if the synapse is not reactivated (so after d seconds, e = 0.5^d).
Learning rate: eta = 0.1.
Prediction error carried by dopamine: delta = +0.8 (unless stated otherwise).
Three-factor update at the moment dopamine arrives: delta_w = eta * e * delta.

Setup

1) Draw one postsynaptic neuron with three input synapses labeled S1, S2, S3.
2) Draw a dopaminergic axon passing over all three (volume transmission: it cannot single any of them out).

Procedure

Part A - One synapse, varying delay

1) S1 is active once at t = 0. Dopamine arrives at t = d.
2) Compute e(d) and delta_w for d = 0, 1, 2, 3, 5, and 10 s. Enter them in a table.
3) Mark the delay beyond which delta_w is below 0.01 (effectively no learning).

Part B - Three synapses, one outcome

4) S1 is active at t = -1 s, S2 at t = -4 s, S3 at t = -8 s. Dopamine arrives at t = 0.
5) Compute the three traces at t = 0 and the three updates.
6) Express the result as a ratio delta_w(S1) : delta_w(S2) : delta_w(S3).

Part C - Negative error

7) Repeat Part B with delta = -0.6 (outcome worse than expected). Which synapses weaken most?

Part D - A trace that never decays

8) Suppose the trace stayed at 1.0 forever. Recompute Part B.
9) Now suppose 50 unrelated synapses were each active at some point during the previous hour. What happens to them when dopamine arrives?
10) Write 8-12 sentences: what does the decay constant buy the organism, what does it cost, and why can the dopaminergic axon be so unselective without destroying specificity?

Optional code extension (Python, about 15 lines)

Write a loop that, for a synapse active at t = 0, computes e(t) every 0.5 s with exponential decay, and prints delta_w for dopamine arrival at each step. Then change the half-life to 3 s and to 0.3 s and compare how far the organism can "reach back" to credit an action.

Lab 3.3 - Credit Assignment Under Uncertainty (a dice-driven learner)

Purpose

To experience credit assignment as the brain faces it: no labels, noisy outcomes, overlapping activity, and a single global signal that cannot name the responsible synapse.

Given model

Four synapses, S1 to S4, each with starting strength w = 0.
On each trial every synapse is independently active with probability 1/2 (die roll of 4, 5, or 6 = active).
Hidden rule (the "truth," which the learner never sees): S1 is the causal synapse. If S1 is active, reward occurs unless the outcome die shows 1. If S1 is inactive, reward occurs only if the outcome die shows 6.
The learner's prediction is fixed at V = 0.5, so delta = r - 0.5 = +0.5 after reward and -0.5 after no reward.
Three-factor update, applied to every synapse that was active on the trial: w = w + 0.1 * delta. Inactive synapses have no trace and do not change.

Procedure

1) Draw the four synapses converging on one neuron, with a dopaminergic axon broadcasting to all four.
2) Make a table with columns: trial, roll S1-S4, active set, outcome roll, reward, delta, w1, w2, w3, w4.
3) Run 12 trials. Either roll your own dice, or use the pre-rolled sequence below so your results can be compared with the class.
4) After each trial update only the active synapses.
5) At the end, rank the four synapses by strength.

Pre-rolled sequence (activity rolls S1, S2, S3, S4; then outcome roll):
trial 1: 1, 4, 1, 4; outcome 2
trial 2: 6, 1, 2, 1; outcome 5
trial 3: 2, 6, 2, 2; outcome 6
trial 4: 1, 6, 1, 2; outcome 2
trial 5: 5, 6, 4, 5; outcome 4
trial 6: 2, 5, 1, 1; outcome 1
trial 7: 4, 3, 2, 2; outcome 3
trial 8: 4, 3, 5, 2; outcome 2
trial 9: 6, 2, 4, 5; outcome 6
trial 10: 5, 1, 1, 2; outcome 3
trial 11: 2, 2, 2, 4; outcome 1
trial 12: 1, 6, 5, 4; outcome 6

Check: with this sequence, after 12 trials you should have w1 = 0.30, w2 = 0.00, w3 = 0.20, w4 = 0.05.

6) Notice that S3 ended with substantial strength although it has no causal role. Find the trials on which it was rewarded, and check whether S1 was also active on those trials.
7) Write 8-12 sentences: did the learner find the causal synapse? Why did S3 get credit it did not deserve? Is that a bug, and what would happen to S3 over 100 more trials? Relate your answer to the chapter's statement that "Bias here does not mean error."

Optional code extension (Python, about 25 lines)

Simulate the same model for 1,000 trials with random rolls and plot the four strengths over time. Then make the outcome noisier (reward with probability 0.6 when S1 is active, 0.4 when inactive) and see how many trials it takes for S1 to separate from the others. The brain lives at the noisy end of this spectrum.

Lab 3.4 - Gated Versus Ungated Plasticity (why activity alone should not decide learning)

Purpose

To make the chapter's design principle concrete: if synapses changed whenever neurons fired, with no neuromodulatory gate, the system would be unstable.

Given parameters

One synapse between neuron A and neuron B, starting strength w = 0.2.
Over 10 trials, A and B happen to fire together on every trial (a coincidence driven by shared input noise, not by any useful relationship).
Ungated (Hebbian) rule: whenever A and B fire together, w = w + 0.1.
Gated (three-factor) rule: w = w + 0.1 * delta, where delta is the prediction error on that trial.
Prediction errors for the 10 trials: 0, 0, 0, +0.5, 0, 0, 0, 0, -0.5, 0.

Procedure

Part A - Coincidence without consequence

1) Run the 10 trials under the ungated rule. Record w after each trial.
2) Run the same 10 trials under the gated rule. Record w after each trial.
3) Compare the two final strengths.

Part B - A real relationship

4) Repeat both rules with prediction errors of +0.5 on every trial (the coactivity now reliably precedes success).
5) Compare the final strengths again.

Part C - Scaling up

6) Suppose the neuron has 5,000 synapses and, by chance, 100 of them show coincident activity on any given trial. Under the ungated rule, how many synapses change per trial? Over 1,000 trials? Under the gated rule with delta = 0 on 90 percent of trials?

7) Write 8-12 sentences: what does gating protect against, what does it cost (think of a genuinely useful coincidence that occurs when delta happens to be zero), and how does this relate to the chapter's claim that "nothing changes unless the context says it should"?

Lab 3.5 - Neuromodulators as Design Primitives (comparison table and concept map)

Purpose

To integrate the chapter's four modulatory systems and its argument about artificial intelligence into one explicit comparison.

Procedure

Part A - Comparison table

1) Make a table with four rows: dopamine, norepinephrine, serotonin, acetylcholine.
2) Add six columns: source region (from the reference table in the Student Notes), what context it signals, typical timescale, effect on plasticity, effect on ongoing processing, and the nearest artificial-learning analogue (for example: reward prediction error term, exploration temperature, learning-rate schedule, attention weighting).
3) Fill every cell with a short concrete statement. Mark any cell where you are guessing.

Part B - Concept map

4) Draw a circuit: a sensory input, a decision neuron, an action, and an outcome.
5) Add a dopaminergic neuron that compares outcome with prediction and broadcasts delta back over the circuit.
6) Add eligibility traces at the synapses as small clocks that start when the synapse is used.
7) Add one gate symbol at each synapse: it opens only when delta is nonzero and the clock has not run out.
8) Add a norepinephrine arrow that widens the gate under surprise, and an acetylcholine arrow that selects encoding versus retrieval.
9) Write an 8-12 sentence caption that tells the story of one trial through your diagram.

Part C - Design memo

10) In 10-14 sentences, argue for or against adding a neuromodulation-like layer to an artificial agent that must learn in an open-ended environment with delayed, noisy feedback. Address stability, credit assignment, and what you would lose compared with backpropagation. Use at least two of the chapter's four implications for AI.